Ex No: 13 Case Study on Limitations of Low-Code Platforms for a Complex Enterprise Use Case

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1. Aim

To study where a low-code platform such as Zoho Creator is a good fit, and where it is limited, for an enterprise-scale hospital appointment management system.

2. Procedure

1. **Case.** A multi-specialty hospital chain uses the appointment application for booking, reports, a command-center page and workflow automation, and now also needs electronic health records, insurance claims integration and queue optimisation across many branches.
2. **Where low-code helps:** forms and fields, reports, dashboards, approvals, blueprints, scheduled reminders, role-based access and quick changes such as adding a new department.
3. **Limitations identified:**
    - **Complex business logic:** advanced rules such as multi-branch doctor-roster optimisation or insurance eligibility checks are hard to express in drag-and-drop workflows.
    - **Scalability and performance:** very large volumes of records and heavy batch jobs may reach platform limits.
    - **Vendor dependency:** all forms, workflows and reports depend on the platform and its pricing.
    - **Integration depth:** complex legacy hospital systems often need custom connectors.
    - **Security and compliance:** health data may need stricter controls and audit than the platform offers by default.
    - **Customisation limits:** highly specialised user interfaces or algorithms may not be possible.
4. **Recommended hybrid architecture:** Layer 1 — low-code application for forms, reports and workflows; Layer 2 — pro-code extensions (custom functions, microservices) for complex logic; Layer 3 — core hospital systems (EHR, billing)

connected by APIs.

[Insert diagram: Hybrid architecture]

3. Result

Low-code is well suited to standard appointment needs, while demanding requirements in logic, integration, security and scale need pro-code extensions connected to core hospital systems. A hybrid architecture combines the speed of low-code with the power of pro-code.